

Student: Jonatan Rabjerg**Instructor:** Andreas Basse-O'Connor**Assignment:** MyLab Math - Uge 11**Date:** 1/15/24**Course:** Calculus Beta

Find the general solution of the given equation.

$$y'' + 36y = 0$$

$y = e^{rx}$ is a solution to the second-order homogeneous differential equation of the form $ay'' + by' + cy = 0$ if and only if r is a solution to $ar^2 + br + c = 0$. The equation $ar^2 + br + c = 0$ is called the auxiliary equation to $ay'' + by' + cy = 0$.

Write the auxiliary equation for the given differential equation.

$$r^2 + 36 = 0$$

Find the roots of the auxiliary equation to determine the form of the general solution to $y'' + 36y = 0$.

$$r^2 + 36 = 0$$

$$r^2 = -36$$

Subtract 36.

$$r = 6i \text{ or } -6i$$

Solve.

The auxiliary equation has two complex roots. Therefore, the general solution to the given differential equation is of the form shown below for $r_1 = \alpha + i\beta$ and $r_2 = \alpha - i\beta$.

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

In this case, where $r_1 = 6i$ and $r_2 = -6i$, $\alpha = 0$ and $\beta = 6$.

Write the general solution to $y'' + 36y = 0$, noting that $e^{0x} = 1$.

$$y = c_1 \cos 6x + c_2 \sin 6x$$